Fiscal Year:	FY 2026	**Task Last Updated:**	FY 04/21/2026
PI Name:	Davis, Catherine M. Ph.D.		
Project Title:	Neurobehavioral Changes Following Stressors and Radiation (NeuroSTAR): Predicting Mission Impacts from Analogous Human and Rodent Endpoints		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	catherine.davis-takacs@usuhs.edu	**Fax:**	FY 301-400-4023
PI Organization Type:	NON-PROFIT	**Phone:**	301-400-4596
Organization Name:	Henry M. Jackson Foundation		
PI Address 1:	6720-A Rockledge Dr.		
PI Address 2:			
PI Web Page:			
City:	Bethesda	**State:**	MD
Zip Code:	20817-1891	**Congressional District:**	8
Comments:	Campus address (Jan 2022): Department of Pharmacology and Molecular Therapeutics, Uniformed Services University of the Health Sciences, 4301 Jones Bridge Road, Bethesda, MD 20814. NOTE: PI formerly at Johns Hopkins University; moved to Henry M. Jackson Foundation for the Advancement of Military Medicine in fall 2020.		
Project Type:	Ground	**Solicitation / Funding Source:**	2024 HERO 80JSC024NA001-FLAGSHIP, OMNIBUS, and NSCOR Crew Health: Topics in Human Health Countermeasures, Behavioral Performance, and Space Radiation and NASA Human Research Program Omnibus Opportunity
Start Date:	02/28/2025	**End Date:**	02/27/2030
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Period of performance (POP) changed to 02/28/2025 - 02/27/2030 per J. McFather/SR. Original POP was 10/01/2024 - 09/30/2029 (Ed., 2/23/26).		
Key Personnel Changes/Previous PI:			

COI Name (Institution):	Basner, Mathias Ph.D. (University of Pennsylvania) Blackwell, Ashley Ph.D. (University of Nevada, Las Vegas) Gur, Ruben Ph.D. (University of Pennsylvania) Jones, Christopher Ph.D. (University of Pennsylvania) Mao, Xiao Wen Ph.D. (Loma Linda University) Mason, Christopher Ph.D. (Weill Medical College of Cornell University) Roalf, David Ph.D. (University of Pennsylvania) Roy, Arkaprava Ph.D. (University of Florida, Gainesville) Seidler, Rachael Ph.D. (University of Florida, Gainesville)
Grant/Contract No.:	80NSSC25K7731
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Multiple stressors, including altered gravity, social isolation and confinement, sleep disruption, and chronic cosmic radiation exposure, threaten optimal human performance during spaceflight. While ground-based laboratory and analog studies have explored how spaceflight stressors can individually impair performance, there is a limited understanding of the degree of impairment following exposure to combined stressors. Further, how chronic cosmic radiation exposure might exacerbate these impairments is unknown. Given the limitations with human epidemiological studies for estimating radiation risk to the central nervous system, there is a need to use other spaceflight-relevant stressors in humans and laboratory animals, and cosmic radiation effects in laboratory animals, to predict radiation-induced decrements in operational performance in astronauts. The objective of our proposal is to identify a series of cognitive and sensorimotor tests that are sensitive to different spaceflight stressors, in both humans and rodents, to ultimately predict potential cosmic radiation-induced cognitive and sensorimotor impairments in human performance. We will assess several different cognitive domains and sensorimotor functions with analogous tests in humans and rodents to determine a subset of operationally relevant cognitive domains and sensorimotor functions degraded by exposure to spaceflight stressors. These experiments will generate dose-response functions for these decrements following exposure to spaceflight stressors, including cosmic radiation, and will use these functions to make predictions regarding radiation-induced decrements in operational task performance outcomes. In addition to cognitive and sensorimotor assessments, we will also determine molecular and physiological markers of spaceflight stressor exposure that will be used as biomarkers verifying stressor exposure and functional performance outcomes. This work is significant because it will employ controlled laboratory studies using well-validated cognitive and sensorimotor assays to test human and rodent performance under various spaceflight-relevant stressors, producing dose-response functions reflecting how these stressors impair human performance. Together, these endpoints will comprise a robust human and rodent behavioral and biological data set to be used for modeling and predicting radiation-induced performance decrements in astronauts on exploration class missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work is significant because it will employ controlled laboratory studies using well-validated cognitive and sensorimotor assays to test human and rodent performance under various spaceflight-relevant stressors, producing dose-response functions reflecting how these stressors impair human performance. Together, these endpoints will comprise a robust human and rodent behavioral and biological data set to be used for modeling and predicting radiation-induced performance decrements in astronauts on exploration class missions.
Task Progress:	This was the first year of the project, so progress is limited. The study team received all human and rodent protocol approvals, and plans are underway for running the first human and rodent subjects this coming year.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Articles in Peer-reviewed Journals	Palmer GA, Mathyk BA, Jones J, Stocks BT, Wolpe PR, Wotring V, Mason CE, Cohen J, Karouia F. "Reproductive biomedicine in space: Implications for gametogenesis, fertility and ethical considerations in the era of commercial spaceflight." Reprod Biomed Online. 2026 Mar;52(3):105431. Review. http://dx.doi.org/10.1016/j.rbmo.2025.105431 ; PubMed PMID: 41638929 , Mar-2026